Squares Whose Equal-Length Halves Differ by a Square

Qichao Wang
Hebei University of Technology
Daoyu Dong
University of Electronic Science and Technology of China

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Abstract

For every positive integer n, we construct a square whose decimal expansion is the concatenation of equal-length positive integers x, y with $y - x = n^2$. This proves the existence conjecture in OEIS A236383. An odd decimal length in one prescribed residue class modulo 22 suffices. Its size is bounded by the number of digits of $4(n+1)^2$ plus 21. We then give an exact finite minimum formula using square roots of n^2 modulo $10^d + 1$, with explicit prime-power root lists and the Chinese remainder theorem.

1 A uniform elementary construction

OEIS A236383 [1] defines $a(n)$ as the least positive k such that k^2 concatenates two positive equal-length integers x, y with $y = x + n^2$. Thus, for some $d \geq 1$,

$$k^2 = 10^d x + y = (10^d + 1)x + n^2, \qquad 10^{d-1} \leq x \leq y < 10^d. \tag{1}$$

The entry conjectures existence for every $n \geq 1$.

Lemma 1. *For every positive odd d, the integer $C_d = (10^d + 1)/11$ satisfies $C_d \equiv d \pmod{11}$.*

Proof. Since d is odd, the binomial expansion of $(-1 + 11)^d + 1$ is congruent to $11d$ modulo 121. Divide by 11. □

Let D be the number of decimal digits of $4(n+1)^2$, and let d_* be the least integer $d \geq D$ satisfying

$$d \text{ odd}, \qquad 5d + 2n \equiv 0 \pmod{11}. \tag{2}$$

The two conditions specify exactly one residue class modulo 22, so $D \leq d_* \leq D + 21$.

Theorem 1. *Put $d = d_*$, $B = 10^d$, $M = B + 1$, and $C = M/11$. Then*

$$k = n + 5C, \qquad x = \frac{5(2n + 5C)}{11}, \qquad y = x + n^2 \tag{3}$$

are positive integers satisfying (1). In particular $a(n)$ exists at every positive index, and $a(n) \leq n + 5C$.

Proof. Lemma 1 and (2) show that $2n + 5C \equiv 0 \pmod{11}$, proving integrality of x. Moreover,

$$k^2 - n^2 = 5C(2n + 5C) = Mx.$$

It remains to check both length inequalities, which congruence alone would not guarantee. The expression for x is

$$x = \frac{25}{121}M + \frac{10}{11}n.$$

Since $25/121 > 1/10$, it gives $x > B/10 = 10^{d-1}$. Also $B \geq 10^D > 4(n+1)^2$, and therefore

$$\frac{96}{121}M > (n+1)^2 > n^2 + \frac{10}{11}n + 1.$$

Rearranging gives $y = x + n^2 < M - 1 = B$. Thus x, y both have exactly d digits and all conditions in (1) hold. There is no exception when $11 \mid n$: the prescribed residue class for d then simply has $11 \mid d$. □

2 The exact least value at every index

For $1 \le d \le d_*$ put $B_d = 10^d$, $M_d = B_d + 1$, and let

$$R_d(n) = \{0 \le r < M_d : r^2 \equiv n^2 \pmod{M_d}\}.$$

If $B_d - 1 - n^2 < 10^{d-1}$, set $\mathcal{K}_d(n) = \varnothing$. Otherwise define the exact integer endpoints

$$\begin{aligned} L_d(n) &= \left\lceil \sqrt{M_d 10^{d-1} + n^2} \right\rceil, \\ H_d(n) &= \left\lfloor \sqrt{M_d(B_d - 1 - n^2) + n^2} \right\rfloor, \\ \mathcal{K}_d(n) &= R_d(n) \cap [L_d(n), H_d(n)]. \end{aligned}$$

Theorem 2. *For every positive n,*

$$\boxed{a(n) = \min \bigcup_{d=1}^{d_*} \mathcal{K}_d(n).}$$

The union is nonempty and all root lists are given by the finite prime-power expressions below. The evaluator can stop at the first d with $\mathcal{K}_d(n) \neq \varnothing$.

Proof. For fixed d, (1) is equivalent to the displayed congruence and the two bounds on k^2. Any valid k satisfies

$$k^2 \le M_d(B_d - 1 - n^2) + n^2 < B_d^2,$$

so $k < M_d$; no translates of the root residues are missing. Theorem 1 supplies a candidate at $d = d_*$. For any admissible length d, (1) implies that k^2 has $2d$ digits, so k has exactly d digits. A larger d can therefore never give a smaller root k. This also proves the stopping rule and excludes all lengths beyond d_*. □

3 Explicit local root lists

Every M_d is odd. For an odd prime power $p^e \parallel M_d$, write $n = p^v u$ with $p \nmid u$. If $2v \ge e$, the complete list of roots modulo p^e is

$$r = jp^{\lceil e/2 \rceil}, \qquad 0 \le j < p^{\lfloor e/2 \rfloor}. \tag{4}$$

If $2v < e$, the complete list is

$$r = p^v\left(\varepsilon u + jp^{e-2v}\right) \pmod{p^e}, \qquad \varepsilon \in \{-1, 1\}, \quad 0 \le j < p^v. \tag{5}$$

To verify completeness, in the first case $r^2 \equiv 0$ is equivalent to $p^{\lceil e/2 \rceil} \mid r$. In the second case any root must have valuation v; after division by p^{2v}, the equation is $z^2 \equiv u^2 \pmod{p^{e-2v}}$. The factors $z - u, z + u$ cannot both be divisible by the odd prime p, since $p \nmid 2u$. Thus $z \equiv u$ or $-u$ modulo that prime power; its lifts give precisely (5). The Chinese remainder theorem combines these local lists to form all of $R_d(n)$. This avoids a scan of consecutive k values.

The formula is an exact finite arithmetic expression involving factorization, explicit root lists, integer square roots and a finite minimum. We do not assert an elementary closed expression without that final minimum. Existence and the length bound are proved for every n, rather than inferred from the finite tests.

4 Independent verification

The exact root evaluator reproduces all 245 official b-file terms. An independent exhaustive scan of $1 \le k \le 1000000$ finds 659 attained indices and their minima; all agree with the evaluator, including all 245 published indices. Further checks compare the prime-power and composite root lists with direct residue enumeration. The explicit construction is checked through $n = 1000$, at powers $11^1, \ldots, 11^{20}$, and at a 101-digit index. The accompanying 300-term table extends the official published range. These checks validate the implementations of the all-index results above.

References

[1] The OEIS Foundation Inc., A236383 and its 245-term b-file, accessed October 26, 2026.